

Linux Programming: Assignment-1:

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ROLL NO- 04

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CLASS- CYBER SECURITY

SECTION- C

1. What is Linux Operating System (OS)? List three pros and cons of it.

Ans:

Linux is a multi-operating system with a command line interface and a graphical user interface. It consists of a kernel, shell, file system, and applications it is popular due to its open source nature, where user can freely modify their software

Pros:

The user can customize the software

It is open and free source

It has high privacy and security, and is stable to use

Cons:

Linux has gaming limitations, as many modern games are designed for the windows os

Linux faces compatibility issues with a few new devices

It has limited availability of certain applications, which forces users to rely on other alternatives

2. Differentiate between Linux, Mac, Android, and Windows OS with at least six unique features.

Ans:

Linux:

It's open source and free

It lets users do multiple tasks at once, and multiple users can use it simultaneously

It's highly customizable, which gives freedom to the user to modify their software

CLI(command line interface) proficient user can easily get control over the system by running a few commands on the terminal

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It is portable: since Android is Linux-based, it is easy to use on any mobile or desktop

MACOs

Pre-installed productivity and multimedia software

It has strict app security and can't be easily accessed

Apple provides consistent major macOS updates with new features, security improvements

It can be much easier if you have Apple integrated system, which can make file transfer a lot easier

Software Updates and Longevity: Regular macOS updates add new features, improve security, and support older devices longer than many competitors.

WINDOWS

It has extensive software compatibility.

It has many built-in network and collaboration tools

Windows often upgrades its software, which reduces the number of security patches

Windows has a built-in antivirus for users

Comes pre-installed on most PCs, offering widespread adoption, from casual to enterprise use.

Older applications can run on newer Windows versions, preserving legacy software usage.

ANDROID

Mainly made for smartphones and tablets.

Owned and developed by Google.

Highly customizable for both users and manufacturers.

Works seamlessly with Google apps like Gmail, Maps, and Drive.
Available on devices ranging from budget to premium.

Supports a huge variety of apps through the Google Play Store.
Regular updates improve security and add new features.

Open-source nature allows developers to create custom versions (like custom ROMs).

Supports multitasking and modern hardware features like cameras, sensors, and biometrics.

Large global user base, making it easy to find help, tutorials, and accessories.

3. Why is Linux preferred for Mainframe Servers for legacy applications? Give three out-of-the-box technical reasons.

Secure: Linux is less prone to viruses as it has very high privacy and stability

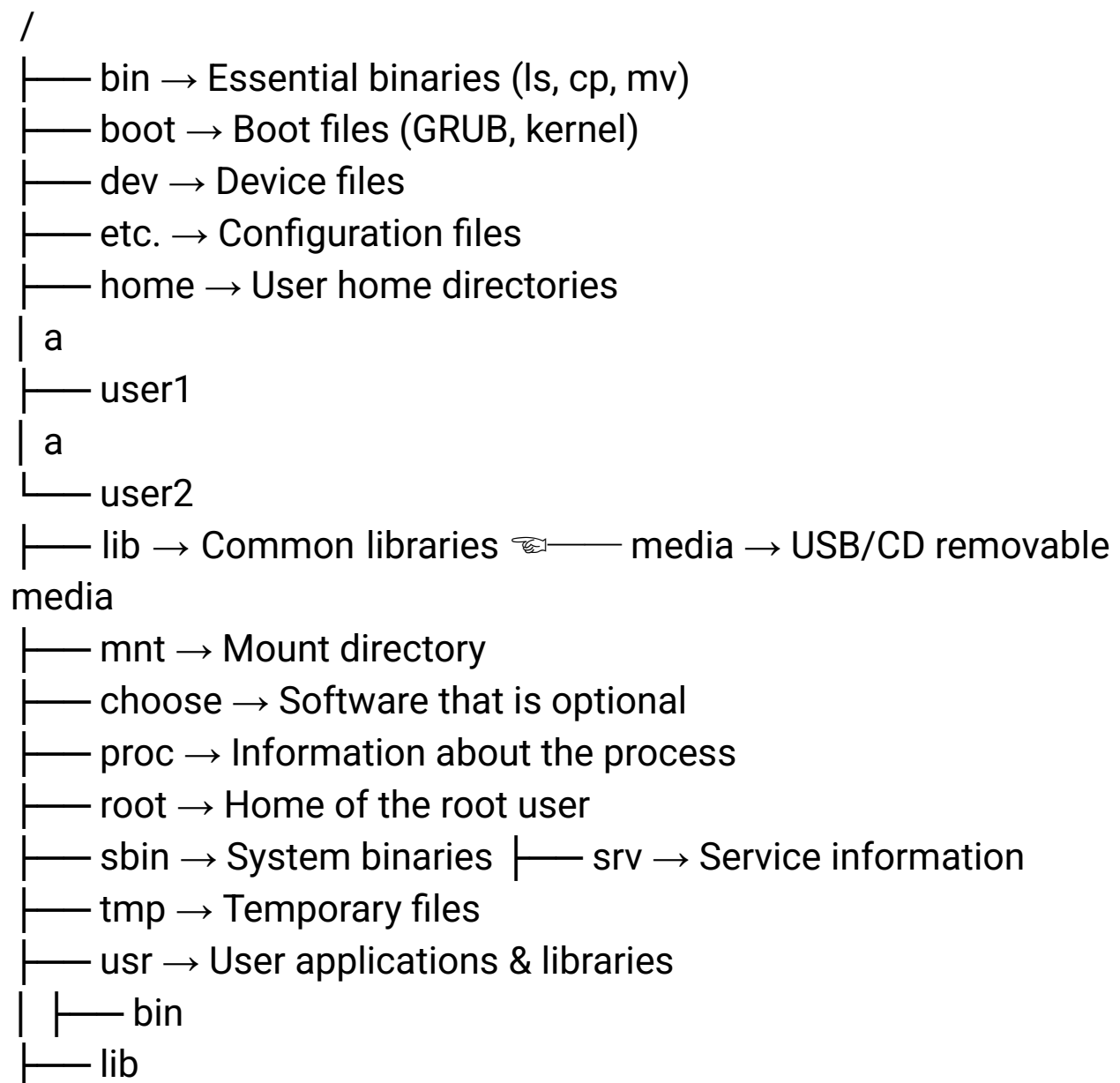
Flexible: since Linux is an open source, users can modify and fit it to their specific needs

Linux handles hardware efficiently, which helps the mainframe to deal with a heavy workload without slowing down

Linux is also cost-friendly, which makes it easy to access, and since it can be used for longer terms, Linux is preferred

4. Explain the structure of the Linux File System with proper diagram. Note: You can use the 'tree' command to find out.

Ans



| └── share
└── var → Variable files (cache, logs)

5. If Linux OS is open-source, how do companies like Red Hat still make money from it? Do a market study and answer properly

- Linux is free and open-source, so anyone can use it without paying.
- Companies like Red Hat don't sell Linux itself but make money through support services.
- They provide enterprise solutions with extra features that help businesses manage large systems.
- Paid training programs, certifications, memberships, and security updates are offered to users and organizations.
- Many also provide cloud services and partnerships with data centers, which bring in extra revenue.
- They sign long-term contracts with big companies to ensure guaranteed support and regular system updates.
- This way, even though Linux is free, these companies earn money through services, solutions, and support

6. Write the command to display today's date and time (i.e., current System time)

Ans:

Command:

date

Return:

Sun Sep 21 19:05:42 IST 2025

7. Which command is used to check how long the system has been running?

Ans:

command : Uptime:13:59:33 up 1 hour, 23 minutes, 2 users,
load average: 0.45, 0.30, 0.20

8. Which command is used to check how long the system has been running?

Ans:

shutdown -h : Safely shuts down the system and powers off.

Halt: Stops the system immediately, but may not power it off properly.

9. Compare init 0 and shutdown -h. Which is safer? Why?

Ans

Init 0: shuts down all processes and services at runlevel

Does not provide any warning before the system goes down

Shutdown -h : halts the system in a controlled manner

Safely shuts down all the file system processes and powers off

shutdown -h is more preferred as it provides a warning before the system shuts down and reduces the risk of data corruption

10. A system administrator accidentally powers off a Server machine without shutting it down properly. What problems can occur with the said Server?

Ans

- Data loss can happen if there are any active files or processes during shutdown
- Recent updates or changes may not be saved
- System recovery or restore processes may have been interrupted
- The server may have trouble booting up next time
- Electronic components may sustain a few damages due to a sudden shutdown
- A sudden shutdown can interrupt the backup process
- Servers may need more time to recover, restart
- Services will suddenly stop, which will disconnect the client and users

Brainstorming

a) As Linux Kernel is open-source, can we build our own operating system?

Yes, it is possible to create your own operating system using the Linux Kernel. The kernel is the main component that manages the hardware and software functions of a computer. Since Linux is open source, users have the freedom to modify it by adding their own software, tools, and designs, and removing parts they don't need to make a fully customized OS. Many popular Linux distributions, like Ubuntu, are created this way; they usually rely on Linux but have their own features. This flexibility allows us to make Linux an excellent choice for anyone wanting to build their own operating system.

b) In order to do that, what are the stoppers, hurdles, and challenges?

Technical background: understanding of operating system , kernel functions, drivers, and system calls is essential.

Programming knowledge: Advanced knowledge of C, along with scripting skills, is required to develop components.

Hardware compatibility: Ensuring that the OS functions reliably across various hardware platforms can be complicated and might need few experts to solve so.

Time and effort: Developing a stable, user-friendly system is time-demanding and lots of consistent effort.

Debugging: Errors and bugs can make the system troublesome, and resolving them often is challenging

Security: The OS must be safeguarded against malware and other threats.

c) Is anyone in India working on this field? Find at-least three to four engineers.

Several Indian engineers have made contributions to Linux:

Ashish Gulhati – Open-source advocate and entrepreneur

Atul Chitnis – Organizer of FOSS.IN

Suparna Bhattacharya – Distinguished Technologist at Hewlett Packard Labs

THANK YOU